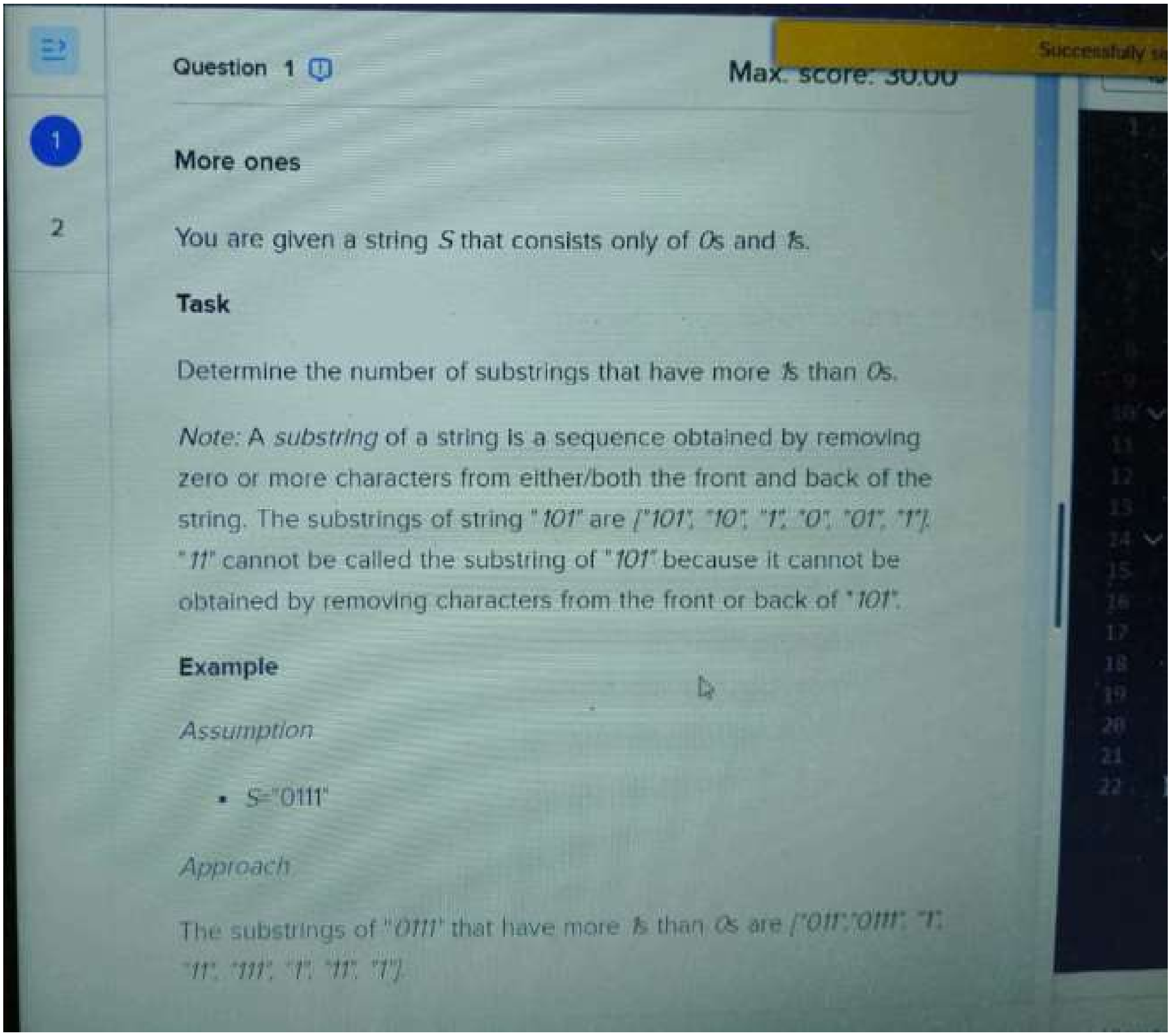
Google - 3

1. **MAX ONES**



Question 1

Max. score: 30.00

More ones

You are given a string *S* that consists only of *0*s and *1*s.

Task

Determine the number of substrings that have more *1*s than *0*s.

Note: A *substring* of a string is a sequence obtained by removing zero or more characters from either/both the front and back of the string. The substrings of string "*101*" are ["*101*", "*10*", "*1*", "*0*", "*01*", "*1*"]. "*11*" cannot be called the substring of "*101*" because it cannot be obtained by removing characters from the front or back of "*101*".

Example

Assumption

- *S*="0111"

Approach

The substrings of "*0111*" that have more *1*s than *0*s are ["*011*","*0111*", "*1*", "*11*", "*111*", "*1*", "*11*", "*1*"]

Approach

The substrings of "*0111*" that have more *1*s than *0*s are *["011","0111", "1", "11", "111", "1", "11", "1"]*.

Therefore, the answer is 8 substrings.

Function description

Complete the *solve* function provided in the editor. This function takes the following parameter and returns the count of substrings that has more *1*s than *0*s.

- *S*: Represents a string denoting the input

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains a single integer *T* that denotes the number of test cases. *T* also denotes the number of times you have to run the *solve* function on a different set of inputs.
- The next *T* lines contain a string denoting S.

Output format

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Successfully signed in as 906979…

Constraints $1 \le T \le 10$

$1 \le |S| \le 10^6$

S *consists of only* '1' *and* '0'

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
2
1010
0111
```

Sample output

```
3
8
```

Explanation

Test Case 1

S = "1010"

The substrings that have more '1' than '0' are ["1" , "1", "101"]. Hence answer is 3

Next Question >

View test cases ^

2. The Maximum Strength

Question 2

Max. score: 30.00

The maximum strength

You are given an array A of size N and Q queries.In each query, you are given an integer M and you have to find the maximum possible strength of the array A. For calculating the strength of the array you can pick any subsequence of the array and add all elements of this subsequence modulo M.

The subsequence of an array can be obtained by erasing some (possibly zero) elements from the array. You can erase any elements, not necessarily going successively. The remaining elements preserve their order. For example, for the array [5,3,1,2,4] the following arrays are subsequences, [3], [5,3,1,2,4], [5,1,4], but the array [1,3] is not.

If you select K elements having index $i_1, i_2, i_3, \ldots i_k$ where $1 \le i_1 < i_2 < i_3 < \ldots i_k \le n$, then Strength = $(A[i_1] + A[i_2] + A[i_3] + \ldots A[i_k])\%M$.

Input format

- The first line contains an integer N representing the size of the array.
- The next line contains N space-separated integers A denoting

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... the array [1,3] is not.

If you select K elements having index $i_1, i_2, i_3, \ldots i_k$ where $1 \leq i_1 < i_2 < i_3 < \ldots i_k \leq n$, then Strength = $(A[i_1] + A[i_2] + A[i_3] + \ldots A[i_k])\%M$.

Input format

- The first line contains an integer N representing the size of the array.
- The next line contains N space-separated integers A_i denoting the elements of the array.
- The next line contains an integer Q representing the number of queries.
- Next Q lines contain an integer M.

Output format

For each query, print the maximum possible strength of the array.

Constraints $1 \leq N \leq 38$

$1 < M, A_i < 10^9$

Output format

For each query, print the maximum possible strength of the array.

Constraints $1 \leq N \leq 38$

$1 \leq M, A_i \leq 10^9$

$1 \leq Q \leq 5$

Sample input

```
3
2 2 3
1
5
```

Sample output

```
4
```

Explanation

In the given test case-

A=[2, 2, 3]

M=5

Explanation

In the given test case-

A=[2, 2, 3]

M=5

In total you have 7 possible way to pick the subsequence, but [2,3] would give you the maximum strength.

so ans=(2+2)%5=4

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 5.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, JavaScript(Node.js), Julia, Kotlin

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3. Queries and Array

Question 1 Max. score: 30.00

Queries and array

You are given three arrays A, B, and C of length N.

For each index i, $A[i]$ is greater than or equal to $B[i]$.

You can perform the given operation on this array A as many times as you want. In each operation:

- Pick any two indices i and j and add 1 to $A[i]$ and subtract 1 from $A[j]$. This operation can only be done only if $A[j]$ is currently greater than $B[j]$. The cost of this single operation is $C[j]$.

Note: 1-based indexing is used.

Task

If you perform the operations optimally, determine the maximum possible minimum number present in the final array A and also find the minimum cost required to make this possible. As this cost can be very large, you are supposed to find it modulo 10^9+7.

Example

Assumptions

- $N = 5$
- $A = [1,2,2,3,3]$
- $B = [1,2,1,2,1]$
- $C = [1,2,1,4,3]$

Approach

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Approach

- The only optimal approach is to choose index *1* as *i* and index *5* as *j* and perform the operation on them.
- The array *A* after that will be *[2, 2, 2, 3, 2]*.
- The maximum possible minimum number in array *A* is *2*. The minimum cost required to do so is *C[5] = 3*.
- So, the required answer is *[2, 3]*.

Function description

Complete the *solve* function provided in the editor. The function takes the following *4* parameters and returns the required answer:

- *N*: Represents an integer denoting the size of the *arrays A, B, and C*
- *A*: Represents the array of integers *A*
- *B*: Represents the array of integers *B*
- *C*: Represents the array of integers C

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains an integer *T* denoting the number of test cases. *T* also denotes the number of times you have to run the *solve* function on a different set of inputs.
- For each test case:
  - The first line contains a single integer *N* denoting the size of the array.
  - The second line contains *N* space-separated integers denoting the elements of the array *A*.
  - The third line contains *N* space-separated integers denoting the elements

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0 / 2 Completed

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
4
1 99 2 99
1 97 2 98
1 5 2 2
```

Sample output

```
3 12
```

Explanation

The first line contains the number of test cases, $T = 1$.

For test case 1

Given

- $N = 4$
- $A = [1, 99, 2, 99]$
- $B = [1, 97, 2, 98]$
- $C = [1, 5, 2, 2]$

Approach

- Choose $i = 1$, $j = 2$ two times and subtract 2 from $2nd$ index element and add 2 in $1st$ index with cost $5*2 = 10$
- Choose $i = 3$, $j = 4$, subtract 1 from $4th$ index element and add 1 in $2nd$ index element with cost 2.
- After that, the final array A will be $[3, 97, 3, 98]$. You cannot increase the value of the minimum element present in array A any more
- The total cost is 12 and the maximum possible minimum value is 3. Hence, the final answer is equal to $[3, 12]$.

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```cpp
#include<bits/stdc++.h>
using namespace std;

vector<int> solve (int N, vector<int> A, vector<int> B, vector<int> C) {
    // Write your code here

}

int main() {

        ios::sync_with_stdio(0);
        cin.tie(0);
        int T;
        cin >> T;
        for(int t_i = 0; t_i < T; t_i++)
        {
            int N;
            cin >> N;
            vector<int> A(N);
            for(int i_A = 0; i_A < N; i_A++)
            {
                cin >> A[i_A];
            }
            vector<int> B(N);
            for(int i_B = 0; i_B < N; i_B++)
            {
                cin >> B[i_B];
            }
            vector<int> C(N);
            for(int i_C = 0; i_C < N; i_C++)
            {
                cin >> C[i_C];
            }
```

above the **Compile and Test** button).

- The first line contains an integer *T* denoting the number of test cases. *T* also denotes the number of times you have to *run* the *solve* function on a different set of inputs.
- For each test case:
    - The first line contains a single integer *N* denoting the size of the array.
    - The second line contains *N* space-separated integers denoting the elements of the array *A*.
    - The third line contains *N* space-separated integers denoting the elements of the array *B*.
    - The fourth line contains *N* space-separated integers denoting the elements of the array C.

Output Format

For each test case in a new line, print the maximum value of the minimum element present in array *A* and the minimum cost to make it possible modulo 10^9+7, in a space-separated format.

Constraints

$1 \le T \le 10^5$

$1 \le N \le 10^5$

$1 \le A[i], B[i], C[i] \le 10^9$

$B[i] \le A[i]$

Sum of N over all testcases $\le 10^5$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

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4. Maximum Gain Substring

Question 1 Max. score: 30.00

Maximum gain substring

You are given a string *str* consisting of *lowercase* English letters and an array *values* of size *26* where *value[i]* denotes the value of i^{th} letter of English alphabet.

Task

You have to extract the substring such that the profit is *maximum*. Profit of string is calculated as the product of the sum of values of letters present in string and the index value of smallest character (in English alphabets) present in the string.

Note: Index value of letters from *a to z* is *1 to 26* respectively.

Example

Assumptions

- *str* = "vju"
- *values* = [2, 3, 4, 4, 2, 2, 1, 5, 3, 5, 3, 5, 3, 2, 4, 3, 4, 1, 1, 1, 2, 1, 5, 4, 1, 4]

Approach

Approach

For this string *6* substrings are possible:

- For substring "*v*", profit is *(1)*(22) = 22*
- For substring "*j*", profit is *(5)*(10) = 50*
- For substring "*u*", profit is *(2)*(21) = 42*
- For substring "*vj*", profit is *(5+1)*(10) = 60*
- For substring "*ju*", profit is *(5+2)*(10) = 70*
- For substring "*vju*", profit is *(5+2+1)*(10) = 80*

Hence, the maximum profit is *80*.

Function description

Complete the function *solve* provided in the editor. This function takes the following *2* parameters and returns the required answer:

- *str*: Represents the given string
- *values*: Represents the value of 26 English letters from *a* to *z*

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button)

- The first line contains an integer *T* denoting the number of test

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- For each test case:
  - The first line contains string *str* denoting the string.
  - The second line contains 26 space-separated integers denoting the value of letters of the English alphabet.

Output format

For each test case in a new line, print the maximum profit you can earn by extracting a substring.

Constraints

$1 \le T \le 10$
$1 \le |str| \le 10^5$
$1 \le values_i \le 10^9$
String consist of lowercase english letters

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
pwr.d
3 2 1 5 5 1 1 2 5 4 1 1 4 5 2 7
```

Sample output

```
117
```

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Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
pmtd
3 2 1 5 3 1 1 2 5 4 1 1 4 5 2 3 3 3
```

Sample output

```
117
```

Explanation

The first line contains the number of test cases, *T = 1*

For first test case

Given

- *str* = *"pmtd"*
- values = *[3, 2, 1, 5, 3, 1, 1, 2, 5, 4, 1, 1, 4, 5, 2, 3, 3, 3, 5, 2, 5, 5, 2, 5, 3, 5]*

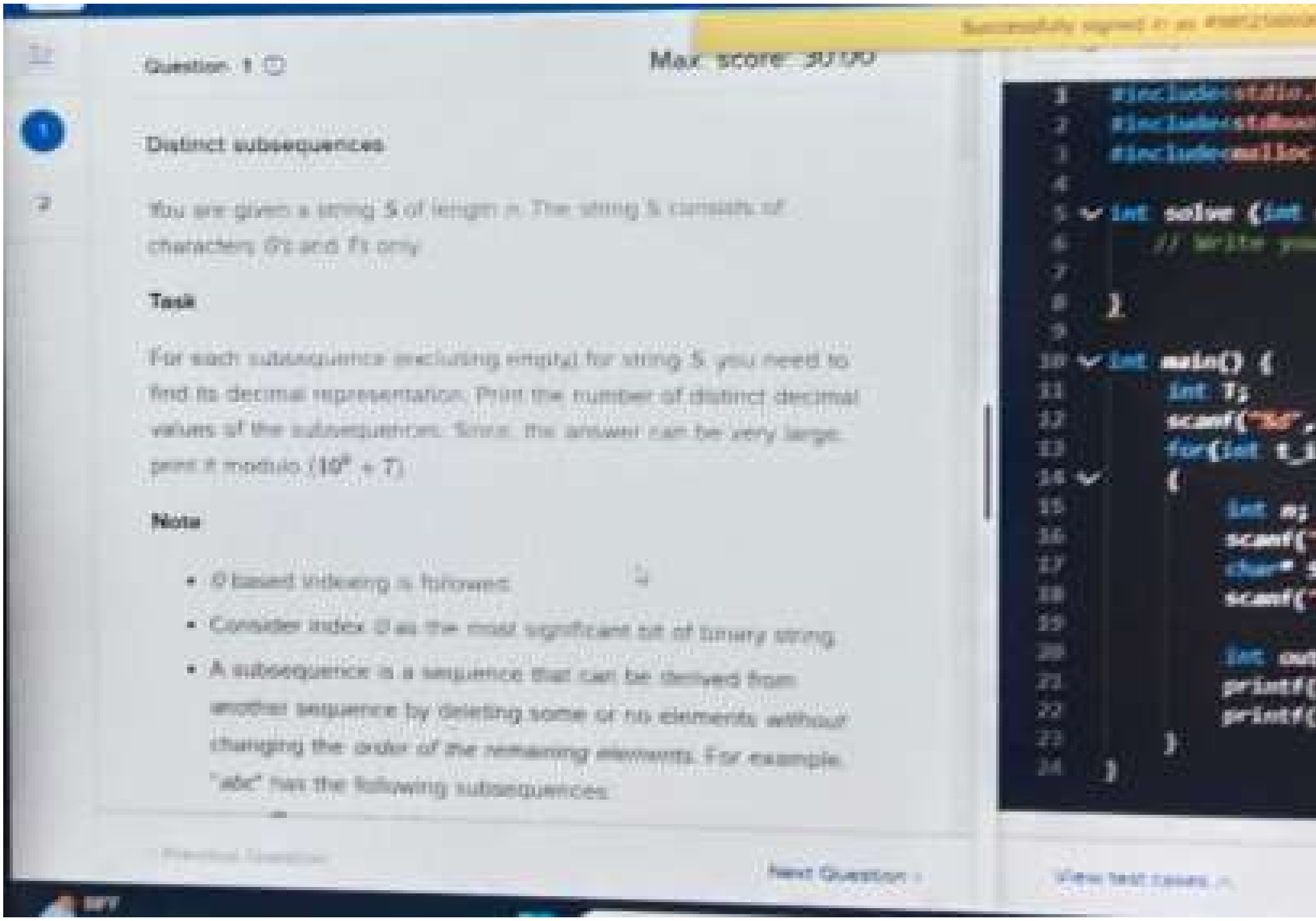
Approach

For this string *10* substrings are possible:

- For substrings of size *1*, which are *"p"*, *"m"*, *"t"* and *"d"* profit is *48, 52, 40* & *20* respectively.
- For substrings of size *2*, which are *"pm"*, *"mt"* and *"td"* profit is *91, 78* & *28* respectively.
- For substrings of size *3*, which are *"pmt"*, *"mtd"* profit is *117, 44* respectively.
- For substrings of size *4*, which are *"pmtd"* profit is *58* .

Hence, the maximum profit is *117*.

5. Distinct Subsequence



6. Median Path

Median path

You are given a tree of n nodes and n - 1 edges. Additionally, you are also given an array C of length n, where C_i $(1 \leq i \leq n)$ denotes the value of node i.

Task

Calculate the sum of medians of every *simple path* of odd length starting from node *1*. (You should add median of node values, that is C_i)

Notes

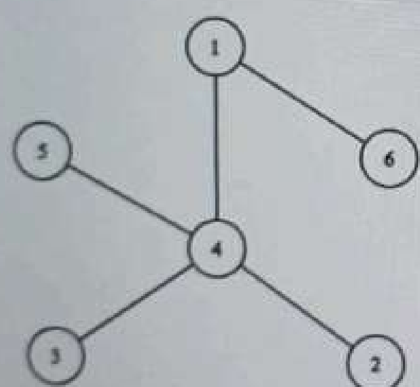
- *1* based indexing is followed.
- The median is the middle number in a sorted (ascending or descending) list of numbers.
- In graph theory, a simple path is a path in a graph that does not have repeating vertices
- A tree is an undirected graph in which any two vertices are connected by exactly one path, or equivalently a connected acyclic undirected graph.

- The median is the middle number in a sorted (ascending or descending) list of numbers.
- In graph theory, a simple path is a path in a graph that does not have repeating vertices
- A tree is an undirected graph in which any two vertices are connected by exactly one path, or equivalently a connected acyclic undirected graph.

Example

Assumptions

- $n = 6$
- $C = [1, 2, 4, 3, 1, 5]$
- Tree:



Approach

All paths of odd length and their median are:

- C_i value in path from *1* to *1* is $\{node_1 = 1\}$ and the median is *1*.
- C_i values in path from *1* to *2* are $\{node_1 = 1, node_4 = 3, node_2 = 2\}$ and their median is *2*.
- C_i values in path from *1* to *3* are $\{node_1 = 1, node_4 = 3, node_3 = 4\}$ and their median is 3.
- C_i values in path from *1* to *5* are $\{node_1 = 1, node_4 = 3, node_5 = 1\}$ and their median is 1.

Hence, the final answer will be $1 + 2 + 3 + 1 = 7$.

Function description

Complete the *solve* function provided in the editor. This function takes the following *3* parameters and returns an integer:

- *n*: Represents the number of nodes in a tree
- *C*: Represents an array denoting the node values
- *edges*: Represents a 2d array of edges

Input format

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- *n*: Represents the number of nodes in a tree
- *C*: Represents an array denoting the node values
- *edges*: Represents a 2d array of edges

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button)

- The first line contains an integer *T* denoting the number of test cases. *T* also denotes the number of times you have to run the *solve* function on a different set of inputs.
- For each test case:
    - The first line contains an integer *n* denoting the number of nodes.
    - The second line contains *n* space-separated integers representing the node values.
    - Next, *n* - *1* line contains two space-separated integers representing an edge between the nodes.

Output format

For each test case, print the answer in a new line representing the sum of medians of every *simple path* of odd length starting from node *1*.

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Output format

For each test case, print the answer in a new line representing the sum of medians of every *simple path* of odd length starting from node *1*.

Constraints

$1 \le T \le 10$
$1 \le n \le 10^5$
$1 \le C_i \le 10^9$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
2
5
7 6 9 10 1
1 2
2 3
3 4
2 5
6
1 2 4 3 1 5
1 4
```

Sample output

```
20
7
```

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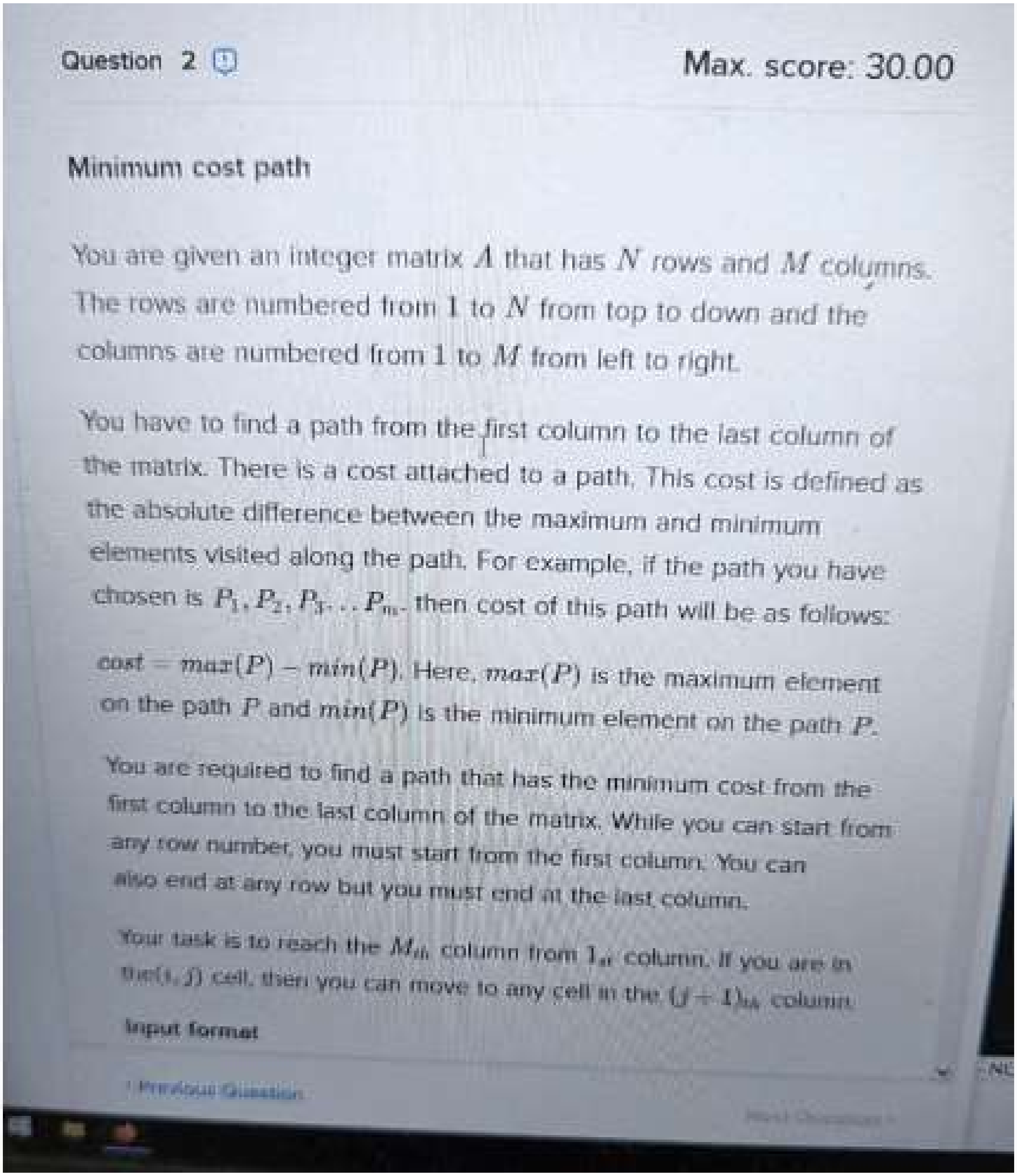
7. Minimum cost Path



Question 2 Max. score: 30.00

Minimum cost path

You are given an integer matrix A that has N rows and M columns. The rows are numbered from 1 to N from top to down and the columns are numbered from 1 to M from left to right.

You have to find a path from the first column to the last column of the matrix. There is a cost attached to a path. This cost is defined as the absolute difference between the maximum and minimum elements visited along the path. For example, if the path you have chosen is $P_1, P_2, P_3 \ldots P_m$, then cost of this path will be as follows:

$cost = max(P) - min(P)$. Here, $max(P)$ is the maximum element on the path P and $min(P)$ is the minimum element on the path P.

You are required to find a path that has the minimum cost from the first column to the last column of the matrix. While you can start from any row number, you must start from the first column. You can also end at any row but you must end at the last column.

Your task is to reach the M_{th} column from 1_{st} column. If you are in the (i, j) cell, then you can move to any cell in the $(j+1)_{th}$ column.

Input format

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Input format

- The first line contains an integer T representing the number of testcases.
- In each test case:
  - The first line contains an integer N representing the number of rows.
  - The second line contains an integer M representing the number of columns.
  - The next N lines contain M space-separated integers representing the elements in matrix A.

Output format

For each test case, print an integer represnting the minimum cost of path in a new line.

Constraints

$1 \leq T \leq 10$
$1 \leq N, M \leq 500$
$0 \leq A_i \leq 10^9$

Sample input

```
2
1
4
1 2 3 4
3
3
1 2 3
1 2 3
2 1 1
```

Sample output

```
3
0
```

Explanation

In the first test case,

There is only one possible path, 1->2->3->4

Cost of this path=4-1=3

In the second test-case,

The best path will be- 1->1->1

Cost=1-1=0

Note:

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1->2->3->4

print the sample output from the provided sample
s run against multiple hidden test cases. Therefore,
hidden test cases to solve the problem statement.

ach input file

estcase passes

ojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8,
js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml,
P, Python, Python 3, Python 3.8, R(RScript), Racket,
TypeScript, Visual Basic

C++14 (g++ 10.3.0) Auto-complete ready!

```cpp
#include<bits/stdc++.h>
using namespace std;

int MinimumCost (int N, int M, vector<vector<int> > A) {
    // Write your code here

}

int main() {

    ios::sync_with_stdio(0);
    cin.tie(0);
    int T;
    cin >> T;
    for(int t_i = 0; t_i < T; t_i++)
    {
        int N;
        cin >> N;
        int M;
        cin >> M;
        vector<vector<int> > A(N, vector<int>(M));
        for (int i_A = 0; i_A < N; i_A++)
        {
            for(int j_A = 0; j_A < M; j_A++)
            {
                cin >> A[i_A][j_A];
            }
        }
```

-NORMAL-

View test cases

Run code